Visiting day talks

10.30-11*, "Short introduction to dispersive PDE -- why do waves spread?" by Mengyi Xie:*

Abstract: In dispersive partial differential equations, waves tend to spread out, as different frequencies propagate at different speeds. This is the driving force that produces dispersive decays. This talk aims to give an intuitive introduction to this mechanism using the linear Schrödinger equation as a model. We will also discuss what happens when nonlinear interactions are added, in which scattering can be observed in many regimes. There are also situations where the dispersion of nonlinear models can be balanced by nonlinear steepening, thereby producing solitons.

11.10-11.40: *"Geometric aspects of the Langlands program" by Joakim Faergeman*

Abstract: We motivate the Langlands program through the Taniyama-Shimura conjecture, which establishes a deep relationship between elliptic curves and modular forms, and whose proof by Andrew Wiles implied Fermat's Last Theorem. We then discuss a geometrization of the Langlands program.

1.45-2.15: *"Integrals and motives" by Alexander Goncharov*

Abstract: The study of integrals of algebraic functions inspired Abel, Riemann and Jacobi to develop the theory of algebraic curves. Theory of motives, conceived by Grothendieck and made precise by Beilinson 40+ years ago, is still largely conjectural. Yet one can already use it to study arbitrary integrals of "algebraic geometric origin". I will explain how these ideas work on the example of the dilogarithm function and its generalizations. During the last 15 years these ideas and methods found their way to Physics, and are now used widely for calculations in collider physics.

2.25-2.55*: "Painleve equations and quantum field theory" by Andrew Neitzke*.

Abstract: The Painleve equations are a family of nonlinear differential equations with remarkable properties, first discovered around 1900, and since then connected to a wide range of subjects in mathematics and physics. I will explain what these equations are, and one of their modern recurrences in conformal field theory.